

The Coherence-Delay Trade-off in Tracking Under Drift

A Cube-Root Law for Finite-Memory Estimation

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Abstract

We study the problem of tracking non-stationary environments under finite memory constraints. When observations are aggregated over time, estimation variance decreases with window size, but delay-induced bias increases as older data becomes less representative under drift. We formalize this trade-off and show that, in the finite-memory setting considered here, it induces a **cube-root scaling law**: the **minimum achievable tracking error** scales as $\zeta^{1/3}$ with the drift rate ζ , with an **optimal effective memory** that scales as $\zeta^{-2/3}$. Empirically, we recover a log-log slope of 0.316 in controlled experiments, closely matching the theoretical prediction.

We further consider action-coupled systems, where observations are influenced by the system’s own interventions. In this setting, apparent improvements in tracking performance may arise from increased control effort rather than improved estimation, leading to a form of feedback-induced masking. To address this, we introduce an **effort-aware coherence diagnostic** that separates genuine adaptation from action-induced alignment. Empirical evaluations on synthetic trackers and real-world datasets show that standard passive drift detectors capture distributional change but do not distinguish these regimes under feedback, while the proposed diagnostic identifies early transitions under intervention with median delay 5.0 steps. These results suggest that tracking under drift is fundamentally constrained by temporal staleness, and that evaluating adaptive systems requires accounting for the interaction between estimation and control.

Keywords: tracking under drift; finite-memory estimation; Wasserstein distance; Sinkhorn divergence; concept drift; online compression; coherence score; feedback masking; coherence-delay trade-off.

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1 Introduction

Modern deployed models operate in environments that are not merely noisy but continuously changing. In recommender systems, adaptive control, and streaming prediction, the system must estimate a time-varying state of the world from finite data while continuing to act on that same world. Those actions can change future observations, so the system faces a joint problem of inference, staleness, and feedback.

A finite-memory estimator benefits from using more observations, because more data reduces statistical uncertainty. But larger windows also make the estimate older. Under drift, that temporal lag produces model staleness: the data used to build the estimate becomes progressively less representative of the present environment. The central question is not whether one can estimate accurately in a stationary regime, but how estimation accuracy is limited once memory and non-stationarity are both unavoidable.

This tension admits a simple form. The total tracking error decomposes into a variance term that falls with window size and a lag-bias term that rises with the drift rate and the effective age of the data. Optimizing this trade-off yields a cube-root law for the finite-memory estimator class analyzed here: $\mathcal{E}_{\min} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$ with $n^* = (C_K/\zeta)^{2/3}$. This law can be read as a limit on *online compression of a non-stationary process*: memory reduces variance, but it also extends the interval over which a compressed model must remain valid while the world keeps moving.

This finite-memory baseline is also the right reference point for evaluating closed-loop diagnostics. In action-coupled systems, lower observed error need not mean better estimation, because interventions can reduce error by steering the environment toward a stale model. The coherence score $\mathcal{C}$ measures the state-update bottleneck directly, while $\mathcal{C}^E$ discounts alignment that is purchased through intervention rather than adaptation.

In closed-loop settings, raw tracking error is not always enough. A system may appear well aligned with its environment because its actions are pushing the environment toward a stale model rather than because the model remains current. This is the phenomenon of *feedback masking*. In the synthetic control regime studied below, the same effect is called coercive masking. A useful decomposition has three functional roles: a representation of the world ($\mathcal{P}$), an actuation rule ($\mathcal{A}$), and an adaptation operator (Φ). It is motivated by action-effective tracking systems rather than by a claim about information in general.

The decomposition yields two diagnostics. The first is the coherence score $\mathcal{C}$, a bottleneck-sensitive score that combines model fidelity, actuation reproducibility, and adaptation quality. The second is an effort-corrected version, $\mathcal{C}^E$, designed for regimes where low observed error may be purchased through intervention rather than through adaptation. The distinction separates degradation from its cause and controls for action-induced confounding.

Section 6 derives the coherence-delay trade-off and the cube-root scaling law. Section 3 shows that action-effective environments require the update to depend on the current state, the most recent action, and the incoming signal, which motivates the decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$. Sections 6 and 8 introduces $\mathcal{C}$ and $\mathcal{C}^E$. Section 7 validates the trade-off and diagnostics in synthetic, passive, and logged settings.

The cube-root law holds for estimators with finite effective memory in the sliding-

window rate class analyzed here. The decomposition is motivated only for *action-effective* systems; passive observers are a genuine dyadic boundary case. The stability results remain conditional on explicit ISS hypotheses, and the diagnostics describe failure modes rather than every possible adaptive architecture.

2 Mathematical Definitions

2.1 State Spaces and Information State

Definition 2.1 (State space). Let S be a non-empty measurable space (the *signal space*). The *information state space* $\mathcal{I}$ is the space of all probability measures on S equipped with a metric $d_{\mathcal{I}}$ compatible with weak convergence. Unless otherwise specified, $d_{\mathcal{I}}$ denotes the L^2 -Wasserstein distance W_2 .

Definition 2.2 (Potentiality). The *Potentiality* at time t is a measurable map

$$\mathcal{P}(S, t) \in \mathcal{I},$$

representing the system’s current probability measure over possible world states. $\mathcal{P}$ is ε -coherent at time t if $W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \varepsilon$, where $\mathcal{P}^*(S, t)$ denotes the true marginal distribution of the environment at time t .

Remark 2.3 (Computational realization of W_2). The L^2 -Wasserstein distance W_2 is chosen as the metric on $\mathcal{I}$ for its geometric faithfulness: unlike f -divergences such as KL divergence, W_2 inherits the geometry of the underlying signal space S and behaves well under spatial translations of the distribution support, a desirable property when concept drift shifts the support of $\mathcal{P}^*$. In high-dimensional signal spaces, exact computation of W_2 costs $O(n^3 \log n)$ via network-flow solvers. For streaming applications requiring real-time Φ_t updates, the entropic regularization of Cuturi [2013] replaces W_2 with the Sinkhorn divergence S_ε , computable in $O(n^2/\varepsilon^2)$ via parallelizable matrix-scaling iterations [Altschuler et al., 2017]. S_ε converges to W_2^2 as $\varepsilon \rightarrow 0$ and breaks the curse of dimensionality of raw W_2 at the cost of a bias proportional to ε [Genevay et al., 2019]. The theoretical results hold for W_2 ; their computational realization in streaming contexts is demonstrated in the Gaussian-tracker experiments (section 7; code at appendix A).

Definition 2.4 (Actuation). The *Actuation operator* is a measurable map

$$\mathcal{A}_t: \mathcal{I} \rightarrow \mathcal{O},$$

where $\mathcal{O}$ is the output space (distinct from lowercase o , which denotes a specific output element $o \in \mathcal{O}$). $\mathcal{A}_t$ is δ -reproducible if for any $p \in \mathcal{I}$ and any two executions ω, ω' , $d_{\mathcal{O}}(\mathcal{A}_t(p; \omega), \mathcal{A}_t(p; \omega')) \leq \delta$. A deterministic actuation is 0-reproducible.

Definition 2.5 (Convergence operator). The *Convergence operator* is a measurable map

$$\Phi_t: \mathcal{I} \times \mathcal{O} \times S \rightarrow \mathcal{I},$$

mapping the current information state, the most recent actuation, and an incoming signal to an updated information state. Φ_t is *coherence-restoring* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \geq 0$,

$$W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \beta(W_2(\mathcal{P}(S, 0), \mathcal{P}^*(S, 0)), t) + \gamma\left(\sup_{s \leq t} \|u(s)\|\right),$$

where $u(s)$ represents the environmental drift input at time s .

Definition 2.6 (Three-Role System). An information system $I(S, t)$ is a *three-role system* if it is equipped with all three operators $(\mathcal{P}, \mathcal{A}, \Phi)$ satisfying definitions 2.2, 2.4 and 2.5 respectively. The composite information state is

$$I(S, t) := (\mathcal{P}(S, t), \mathcal{A}_t, \Phi_t) \in \mathcal{I} \times \mathcal{F}_{\mathcal{O}} \times \mathcal{F}_{\mathcal{I}}.$$

2.2 ISS Definitions

Definition 2.7 (Class $\mathcal{K}$, $\mathcal{KL}$). A function $\gamma: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is class $\mathcal{K}$ if it is continuous, strictly increasing, and $\gamma(0) = 0$. A function $\beta: \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}_{\geq 0}$ is class $\mathcal{KL}$ if $\beta(\cdot, t) \in \mathcal{K}$ for each t and $\beta(r, \cdot) \rightarrow 0$ as $t \rightarrow \infty$ for each $r > 0$.

Definition 2.8 (Input-to-State Stability). A dynamical system $\dot{x} = f(x, u)$ is *input-to-state stable* (ISS) if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for every initial state $x(0)$, every essentially bounded input u , and all $t \geq 0$,

$$\|x(t)\| \leq \beta(\|x(0)\|, t) + \gamma(\|u\|_{\infty}).$$

3 Why Action Enters the Update in Closed-Loop Tracking

The coherence-delay law in section 6 concerns estimation with finite memory. In open-loop settings, the estimator only needs to combine past state information and new observations. In closed-loop settings, however, the system's own actions alter future observations. The update therefore needs to keep track of what the system did, not only of what it saw.

3.1 External Characterization: Drift-Coherence

Definition 3.1 (Drift-Coherence). An information system $I(S, t)$ is *drift-coherent* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \geq 0$ and all environmental inputs with $\text{ess sup}_{s \leq t} \|u(s)\| \leq M < \infty$:

$$W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \beta(W_2(\mathcal{P}(S, 0), \mathcal{P}^*(S, 0)), t) + \gamma(M).$$

The system maintains bounded tracking error under bounded drift.

Definition 3.2 (Action-Effectiveness). The actuation operator $\mathcal{A}_t$ is *action-effective* in environment $\mathcal{X}$ if there exist distinct outputs $o, o' \in \mathcal{O}$ and a state $s \in S$ such that

$$D_{\text{TV}}\left(P(\cdot \mid s_t = s, o) \parallel P(\cdot \mid s_t = s, o')\right) > 0.$$

If this never occurs, the system is *passive*: its actions do not affect the environmental transition law.

3.2 Belief Update With Feedback

Lemma 3.3 (Belief Update Lemma). *Let $I(S, t)$ be a drift-coherent, action-effective information system. Any update map U whose repeated application preserves drift-coherence in this class must depend on three arguments:*

- (i) *the current internal state $p = \mathcal{P}(S, t)$,*
- (ii) *the most recent action $o = \mathcal{A}_t(\mathcal{P})$, and*
- (iii) *the incoming environmental signal s_{t+1} .*

Each argument cannot be dropped without leaving the class of drift-coherent updates considered here.

Proof. If U omits p , the update becomes memoryless and discards the sufficient summary of the past; under temporally structured drift this prevents bounded tracking error. If U omits o , then in an action-effective environment the update cannot distinguish exogenous drift from changes induced by the system’s own actuation. This is precisely the credit-assignment problem that appears in POMDP filtering and belief update semantics [Kaelbling et al., 1998, Katsuno and Mendelzon, 1992, Bonanno, 2024]. If U omits s_{t+1} , no new environmental evidence enters the state estimate and the gap between $\mathcal{P}$ and $\mathcal{P}^*$ grows with drift. In each case, the drift-coherence bound fails. $\square$

Remark 3.4 (Passive observer systems). When actions do not influence transitions, the update can drop the actuation argument and the system is genuinely dyadic. The three-role decomposition is therefore motivated only for action-effective settings.

Corollary 3.5 (Ternary Update Structure). *For action-effective tracking systems, any drift-coherent update map has the form*

$$U: \mathcal{I} \times \mathcal{O} \times S \rightarrow \mathcal{I}.$$

Any ostensibly smaller description either hides one of these arguments inside a state variable or fails drift-coherence.

Proof. This is immediate from lemma 3.3. $\square$

Theorem 3.6 (Minimal Functional Roles). *Let $I(S, t)$ be a drift-coherent, action-effective information system. Any functional decomposition that preserves corollary 3.5 in this class must realize three roles: a stored state, an action channel, and an adaptation rule that maps the state, action, and new observation to an updated state. The tuple $(\mathcal{P}, \mathcal{A}, \Phi)$ is the explicit decomposition used here.*

Proof. By corollary 3.5, the update requires all three arguments (p, o, s_{t+1}) . A decomposition that omits any one of the corresponding roles must either re-encode it implicitly inside another component or lose the update structure required for drift-coherence. The explicit realization used here is $\mathcal{P}$ for stored state, $\mathcal{A}$ for the action channel, and Φ for the update map. $\square$

4 A Minimal Stability Template for Closed-Loop Tracking

The necessity argument in section 3 identifies the three inputs a closed-loop update must use. To turn that decomposition into a usable diagnostic, we need a stability template under which each component admits a coherence score. We use an ISS-based cascade model for that purpose.

4.1 Hypotheses

The following assumptions are explicit and application-dependent.

- (H-ISS-P) The representation subsystem admits an ISS-Lyapunov function $V_P: \mathcal{I} \rightarrow \mathbb{R}_{\geq 0}$.
- (H-ISS-A) The actuation subsystem admits an ISS-Lyapunov function V_A .
- (H-ISS- Φ) The adaptation subsystem admits an ISS-Lyapunov function V_Φ .
- (H-ISS-Env) The environmental drift input is essentially bounded and evolves slowly enough for Φ_t to remain contractive on the induced state trajectory.

Remark 4.1 (Measure-valued state spaces). When $\mathcal{P}(S, t)$ is measure-valued, the natural potential function is $V_P = \frac{1}{2}W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t))^2$. In Gaussian instances this reduces to an ordinary quadratic tracking error; in more general settings the construction follows the Wasserstein gradient-flow viewpoint [Ambrosio et al., 2005, Mironchenko and Prieur, 2020].

4.2 Main Stability Theorem

Theorem 4.2 (Borromean Cascade Stability). *Under hypotheses (H-ISS-P), (H-ISS-A), (H-ISS- Φ), and (H-ISS-Env), the composite system $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ with feedback structure*

$$\Phi_t \xrightarrow{\text{updates}} \mathcal{P}(S, t) \xrightarrow{\text{drives}} \mathcal{A}_t \xrightarrow{\text{feeds back}} \Phi_t$$

admits the composite Lyapunov functional $V_{\text{total}} := V_P + V_A + V_\Phi$, which is non-increasing along trajectories for bounded environmental inputs. Removing any one of the three roles destroys this guarantee in general.

Proof. The cascade ($\Phi \rightarrow \mathcal{P} \rightarrow \mathcal{A}$) is ISS by repeated application of standard cascade arguments [Sontag, 1989, 1995, Dashkovskiy et al., 2010]. The subsystem Lyapunov functions combine into a composite functional whose cross terms are absorbed by the ISS gains. If any one of the three roles is removed, at least one of the required input channels disappears: without Φ the state is not corrected, without $\mathcal{P}$ the action has no bounded state input, and without $\mathcal{A}$ the feedback loop to adaptation is broken. In each case the monotone decrease of V_{total} is no longer guaranteed. $\square$

Example 4.3 (Concrete Gaussian instantiation). Let $\mathcal{P}^*(S, t) \sim \mathcal{N}(\mu_t, \Sigma)$ with

$$\mu_t = \mu_0 + \int_0^t \eta(s) ds, \quad \|\eta\|_\infty < M.$$

Let $\mathcal{P}(S, t) = \mathcal{N}(\hat{\mu}_t, \Sigma)$ be the Kalman estimate, $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$, and Φ_t the Kalman update. Then $V_P = W_2(\mathcal{P}, \mathcal{P}^*)^2 = \|\hat{\mu}_t - \mu_t\|^2$, together with the standard actuation and innovation terms, yields a concrete instance satisfying the ISS template [Jazwinski, 1970].

5 Diagnostic Failure Modes

The decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ is useful only if it identifies failure regimes that matter empirically. This section records the three diagnostic failure modes used throughout the experiments and the coherence scores.

Proposition 5.1 (Failure Mode 1: Static- $\mathcal{P}$ collapse). *Suppose Φ_t ceases to update $\mathcal{P}$, so $\mathcal{P}(S, t) = \mathcal{P}(S, t_0)$ for all $t > t_0$. If the environment undergoes distributional shift $\mathcal{P}^*(S, t) \neq \mathcal{P}^*(S, t_0)$ for large t , then $W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \rightarrow \infty$ and the expected actuation error $\mathbb{E}[d_{\mathcal{O}}(\mathcal{A}_t(\mathcal{P}), \mathcal{A}_t^*)]$ is unbounded.*

Proof. $\mathcal{P}(S, t) = \mathcal{P}(S, t_0)$ is fixed while $\mathcal{P}^*$ drifts, so $W_2 \rightarrow \infty$ by definition. Lipschitz continuity of $\mathcal{A}_t$ then propagates the divergence to actuation error. $\square$

Remark 5.2 (Interpretation). FM-1 is the stale-model regime: the representation no longer tracks the world, so downstream actions inherit an increasingly obsolete state estimate.

Proposition 5.3 (Failure Mode 2: Non-deterministic- $\mathcal{A}$ divergence). *If $\mathcal{A}_t$ is not δ -reproducible for any $\delta < \infty$, then $H(\mathcal{A}_t \mid \mathcal{P}(S, t)) > 0$ for all t : actuation carries information not determined by the state model. Distributed replicas of the system diverge, and the composite resists auditing and replication.*

Proof. Non- δ -reproducibility implies $\exists \omega \neq \omega'$ with $\mathcal{A}_t(p; \omega) \neq \mathcal{A}_t(p; \omega')$, giving positive conditional entropy. Auditability requires reconstructing actuation from state; this fails when conditional entropy is positive. $\square$

Remark 5.4 (Interpretation). FM-2 is the non-reproducible-action regime: the model may still be accurate, but the realized action can no longer be reconstructed from the represented state, so replicas and audits diverge.

Proposition 5.5 (Failure Mode 3: Absent- Φ monopolization). *Let $\Phi_t \equiv \text{id}$ (no update). In any closed-loop deployment, actuation feedback shifts $\mathcal{P}^*$; since $\mathcal{P}$ cannot track, the distributional mismatch grows at a rate lower-bounded by the rate of environmental response to actuation.*

Proof. The closed-loop case follows from $\mathcal{P}^*(t+1) \leftarrow f(\mathcal{P}^*(t), \mathcal{A}_t(\mathcal{P}(t)))$ with $\mathcal{P}$ fixed, so the gap $W_2(\mathcal{P}, \mathcal{P}^*)$ grows at least as fast as the variation of f in its second argument whenever $\mathcal{A}_t(\mathcal{P}) \neq \mathcal{A}_t^*$. $\square$

Remark 5.6 (Interpretation). FM-3 is the missing-adaptation regime. In passive settings it appears as a failure to track external drift; in closed-loop settings it can amplify the system's own feedback and produce self-reinforcing misalignment.

6 Coherence Diagnostics

The coherence-delay law constrains the representation term of any finite-memory tracking system, but closed-loop performance also depends on whether actions remain reproducible and whether adaptation remains effective. The ISS template of section 4 provides one way to score those three ingredients. This section packages them into a diagnostic built from the minimal decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ used throughout the paper.

Definition 6.1 (Component Coherence Scores). Let $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ be an information system with ISS-Lyapunov functionals V_P, V_A, V_Φ as in section 4. Define the three *component coherence scores* at time t as:

$$\sigma_P(t) := \frac{1}{1 + V_P(t)} = \frac{1}{1 + \frac{1}{2}W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t))^2}, \quad (1)$$

$$\sigma_A(t) := \exp(-H(\mathcal{A}_t \mid \mathcal{P}(S, t))), \quad (2)$$

$$\sigma_\Phi(t) := \frac{1}{1 + V_\Phi(t)}. \quad (3)$$

Each score takes values in $(0, 1]$, equals 1 when the corresponding component is ideal, and approaches 0 as the component degrades.

Definition 6.2 (Coherence Score). The *coherence score* of a system $I(S, t)$ at time t is

$$\mathcal{C}(I(S, t), t) := \min(\sigma_P(t), \sigma_A(t), \sigma_\Phi(t)) \in (0, 1].$$

The min aggregation is the correct choice given the Borromean property (theorem 4.2): system coherence is bounded by the weakest component, and no strength in two components compensates for failure in the third. An arithmetic mean would obscure localized failures; the min preserves the diagnostic identity of the bottleneck.

Remark 6.3 (Bottleneck sensitivity and effort correction). The use of $\min(\sigma_P, \sigma_A, \sigma_\Phi)$ is bottleneck-sensitive: it is a diagnostic of the weakest active subsystem, not a smooth summary of overall system quality. Aggregators such as arithmetic or geometric means may be reported as descriptive summaries, but they are not valid replacements for the structural diagnostic given by definition 6.2.

In action-effective environments, however, high actuation capacity can keep σ_P artificially high by forcing the world toward a stale model. When such feedback masking is possible, a conservative extension is the *effort-corrected score*

$$\mathcal{C}^E(I(S, t), t) := \min(\sigma_P(t)e^{-\lambda E_t/E_0}, \sigma_A(t), \sigma_\Phi(t)),$$

where E_t is any monotone proxy for corrective actuation burden, $E_0 > 0$ is a reference scale, and $\lambda > 0$ controls sensitivity. It discounts apparent improvements in model-world alignment when those gains come from action-induced confounding rather than better state estimation. In the synthetic particle tracker, E_t is the realized coercive effort $\alpha|a_t - s_t^{\text{free}}|$; in deployed systems, action increments, energy expenditure, or control-surface usage can play the same role. For logged recommendation benchmarks, the preferred instantiation is the per-user KL divergence between the current exposure-tag distribution and the healthy random-exposure baseline distribution, aggregated sequentially over time and then batched across users.

Remark 6.4 (Finite-memory interpretation). The scaling law behind proposition 6.7 is not an artefact of the sliding-window estimator alone. The two terms in the bound, $n^{-1/2}$ and $\zeta\tau$, encode the structural trade-off between variance reduction and delay-induced bias that any estimator with finite effective memory must confront: the first term reflects the statistical limit of estimating a distribution from finitely many samples under subgaussian noise, and the second reflects the mismatch induced by drift when the estimator necessarily looks at stale information.

Informally, one may interpret this as a near-minimax phenomenon over the class of trajectories with bounded drift rate ζ : if two admissible trajectories differ only by a perturbation below the resolution induced by finite memory, they are indistinguishable at the level of observations, and any estimator must incur an error of the same order. The estimator used in proposition 6.7 achieves the resulting cube-root scaling up to constants, so the $\zeta^{1/3}$ law should be read as a robust operational regime under the stated regularity assumptions rather than as a peculiarity of a single windowing scheme.

More concretely, any estimator with finite effective memory must incur both a statistical error term of order $n^{-1/2}$ and a drift-induced bias of order ζn , so no choice of estimator can simultaneously eliminate both contributions. This argument applies to estimators with finite effective memory, including sliding-window and exponentially weighted schemes.

Optimizing the balance between these unavoidable contributions yields the cube-root scaling. Any improvement beyond this regime must therefore come from stronger assumptions on the drift process or the observation model, not merely from a different finite-memory estimator.

6.1 Estimable Score

Definition 6.1 requires $\mathcal{P}^*(S, t)$, the true oracle distribution of the environment, to compute $\sigma_P(t)$. In any deployed system, $\mathcal{P}^*$ is unobservable by construction: a system with complete knowledge of $\mathcal{P}^*$ would have nothing to estimate. We therefore introduce an estimable version of σ_P using an empirical approximation of $\mathcal{P}^*$.

Definition 6.5 (Empirical Potentiality Estimator). Let $\{s_{t-n+1}, \dots, s_t\}$ be the n most recent environmental signals observed by the system. The *empirical potentiality estimator* is the empirical measure:

$$\hat{\mathcal{P}}^*(S, t) := \frac{1}{n} \sum_{k=t-n+1}^t \delta_{s_k},$$

where δ_{s_k} is the Dirac measure at observation s_k .

Definition 6.6 (Estimable Score). The *debiased Sinkhorn divergence* between measures μ, ν is

$$S_\varepsilon(\mu, \nu) := T_\varepsilon(\mu, \nu) - \frac{1}{2}T_\varepsilon(\mu, \mu) - \frac{1}{2}T_\varepsilon(\nu, \nu),$$

where T_ε is the raw entropically regularized optimal transport cost. Unlike T_ε , the debiased divergence satisfies $S_\varepsilon(\mu, \mu) = 0$, restoring the metric property and enabling the dimension-independent convergence rate of proposition 6.7.

The *estimable component* $\hat{\sigma}_P$ replaces $\mathcal{P}^*$ with $\hat{\mathcal{P}}^*$ in eq. (1) using S_ε :

$$\hat{\sigma}_P(t) := \frac{1}{1 + \frac{1}{2}S_\varepsilon\left(\mathcal{P}(S, t), \hat{\mathcal{P}}^*(S, t)\right)^2}.$$

The *estimable score* is

$$\hat{\mathcal{C}}(I(S, t), t) := \min(\hat{\sigma}_P(t), \sigma_A(t), \sigma_\Phi(t)).$$

Note that σ_A and σ_Φ remain exactly computable without oracle access: $\sigma_A = e^{-H(\mathcal{A}_t|\mathcal{P})}$ is estimated from replica divergence logs, and $\sigma_\Phi = 1/(1 + V_\Phi)$ from the convergence subsystem's Lyapunov functional.

Proposition 6.7 (Estimation Error Bound for $\hat{\sigma}_P$). *Let $\mathcal{P}^*(S, t)$ be K -subgaussian with drift rate $\zeta = \text{ess sup}_s \|\dot{\mathcal{P}}^*(s)\|_{W_2}$. Let $\hat{\mathcal{P}}^*$ be the sliding-window empirical estimator with window n (definition 6.5), and let τ be the lag between the center of the window and t . Then under the entropic regularization of remark 2.3:*

$$|\sigma_P(t) - \hat{\sigma}_P(t)| \leq \frac{C_K \cdot (n^{-1/2} + \zeta\tau)}{(1 + V_P(t))^2},$$

where C_K depends only on the subgaussian constant K .

Proof. By the triangle inequality for S_ε :

$$|S_\varepsilon(\mathcal{P}, \mathcal{P}^*) - S_\varepsilon(\mathcal{P}, \hat{\mathcal{P}}^*)| \leq S_\varepsilon(\mathcal{P}^*, \hat{\mathcal{P}}^*).$$

Under K -subgaussian $\mathcal{P}^*$, Theorem 3 of [Genevay et al. \[2019\]](#) establishes that the debiased Sinkhorn divergence achieves a dimension-independent parametric rate: $\mathbb{E}[S_\varepsilon(\mathcal{P}^*, \hat{\mathcal{P}}^*)] \leq C_{K,\varepsilon} n^{-1/2}$, where $C_{K,\varepsilon}$ depends on the subgaussian norm K and the regularization parameter ε . (Theorem 2 of the same paper, which proves that the Sinkhorn potentials lie in a Sobolev ball independent of the measures, is the enabling lemma for this rate.) The result holds when $\mathcal{P}^*$ is supported on bounded subsets of $\mathbb{R}^d$ with C^1 cost; for unbounded subgaussian measures, see [Mena and Weed \[2019\]](#) for an improvement. This rate is specific to the debiased S_ε ; the raw cost T_ε retains an entropy bias $T_\varepsilon(\mu, \mu) > 0$ that prevents the dimension-independent convergence from holding. An additional drift error of $\zeta\tau$ arises because $\hat{\mathcal{P}}^*$ is centered τ steps in the past. The bound on $|\sigma_P - \hat{\sigma}_P|$ follows from the Lipschitz continuity of $x \mapsto 1/(1 + \frac{1}{2}x^2)$ with derivative $\leq 1/(1 + V_P)^2$. $\square$

proposition 6.7 grounds $\hat{\mathcal{C}}$: the estimation error decreases as the window n grows and increases proportionally to drift rate ζ . The required window size for target error η is $n = O(C_{K,\varepsilon}^2/\eta^2)$, independent of dimension.

This characterization also yields the tracking limit used throughout the paper. When window lag is accounted for explicitly ($\tau = n/2$ for a centred window), the total estimation error decomposes as a sum of two opposing terms. Minimising over n yields a lower bound on achievable estimation accuracy for estimators with finite effective memory, including sliding-window and exponentially weighted schemes.

Definition 6.8 (Lag-Inclusive Estimation Error). For a sliding window of size n centred at lag $\tau = n/2$, the total expected estimation error is:

$$\mathcal{E}(n) = C_K n^{-1/2} + \frac{1}{2} \zeta n,$$

where the first term is the *variance* (decreasing in n) and the second is the *lag-bias* (increasing in n).

Proposition 6.9 (Optimal Window Under Drift). *Under definition 6.8, the estimation error $\mathcal{E}(n)$ is minimised at the optimal window*

$$n^* = \left(\frac{C_K}{\zeta} \right)^{2/3},$$

yielding the minimum achievable tracking error

$$\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}.$$

Proof. Setting $d\mathcal{E}/dn = 0$: $-\frac{1}{2}C_K n^{-3/2} + \frac{1}{2}\zeta = 0$, giving $n^* = (C_K/\zeta)^{2/3}$. Substituting back: $\mathcal{E}(n^*) = C_K(C_K/\zeta)^{-1/3} + \frac{1}{2}\zeta(C_K/\zeta)^{2/3} = C_K^{2/3}\zeta^{1/3} + \frac{1}{2}C_K^{2/3}\zeta^{1/3} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$. $\square$

Remark 6.10 (Interpretation: a finite-memory floor). Proposition 6.9 establishes that perfect coherence ($V_P = 0$) is *impossible* under continuous drift for estimators with finite effective memory in the sliding-window rate class analyzed here. Every drift-coherent system must carry a residual tracking error at least $\mathcal{E}_{\min} \propto \zeta^{1/3}$. This can be read as a finite-memory limitation: the two sources of error (variance and lag) trade against each other with no joint optimum of zero. The cube-root scaling has a concrete engineering consequence: a fifty-fold increase in drift rate ζ raises the minimum error floor by only $50^{1/3} \approx 3.7\times$, while simultaneously forcing the optimal window n^* to contract by $50^{2/3} \approx 14\times$. The optimal window $n^* = (C_K/\zeta)^{2/3}$ provides a *self-tuning rule*: given a real-time estimate $\hat{\zeta}_t$ of the local drift rate, the system should maintain window size $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$ to operate at the minimum of the error U-curve at every step. This turns the coherence-delay law into a practical self-tuning rule for the windowed estimator class analyzed here. In practice, this implies that improvements in apparent tracking performance must be interpreted relative to this baseline, particularly in systems where actions influence observations.

Remark 6.11 (Practical estimation of V_Φ for black-box Φ_t). The estimable score requires $\sigma_\Phi = 1/(1 + V_\Phi)$, where V_Φ is the ISS-Lyapunov functional for the convergence subsystem. When Φ_t is a black box, with internal dynamics unknown and ISS comparison functions $(\alpha_{1\Phi}, \alpha_{2\Phi}, \alpha_{3\Phi}, \chi_\Phi)$ unavailable, V_Φ admits no closed-form derivation. In such cases, V_Φ is estimated empirically via Counterexample-Guided Inductive Synthesis (CEGIS): a learner proposes a candidate $\hat{V}_\Phi$ from observed state trajectories $\{(\Phi_t(p_i, o_i, s_i), p_i)\}_i$; a verifier checks the dissipation condition $\Delta\hat{V}_\Phi \leq 0$ against sampled trajectories using local Lipschitz bounds; failed verification returns counterexamples to the learner. The resulting $\hat{V}_\Phi$ yields a conservative lower-bound estimate $\hat{\sigma}_\Phi = 1/(1 + \hat{V}_\Phi) \leq \sigma_\Phi$, usable in $\hat{\mathcal{C}}$ for any opaque convergence subsystem.

6.2 Architecture and Behavior Are Not Equivalent

A structural asymmetry in the framework deserves explicit statement.

Drift-coherence implies the need for all three roles. By lemma 3.3 and theorem 3.6, any drift-coherent, action-effective system must implement the three functional roles $(\mathcal{P}, \mathcal{A}, \Phi)$. Drift-coherence is the stronger behavioral condition; the decomposition is its architectural consequence in action-effective settings.

Implementing the decomposition is not enough. A system may implement $(\mathcal{P}, \mathcal{A}, \Phi)$ structurally while failing to satisfy the ISS hypotheses (H-ISS-P/A/ Φ /Env) required for theorem 4.2. The components may be present but improperly tuned, have inadequate bandwidth, or lack sufficient expressiveness.

Proposition 6.12 (The Score Characterizes Component-Level Failure). *Let $I(S, t)$ be a three-role system under hypotheses (H-ISS-P/A/ Φ /Env). Then:*

- (i) $\mathcal{C}(I(S, t), t) = 1$ if and only if $V_P(t) = 0$, $H(\mathcal{A}_t \mid \mathcal{P}) = 0$, and $V_\Phi(t) = 0$ simultaneously.
- (ii) $\mathcal{C}(I(S, t), t) \rightarrow 0$ if and only if at least one of the three failure modes is active: $V_P(t) \rightarrow \infty$ (FM-1), $H(\mathcal{A}_t \mid \mathcal{P}) \rightarrow \infty$ (FM-2), or $V_\Phi(t) \rightarrow \infty$ (FM-3).
- (iii) The identity of $\arg \min(\sigma_P, \sigma_A, \sigma_\Phi)$ names the bottleneck component.

Proof. (i) Each score is a strictly decreasing function of its Lyapunov functional or conditional entropy: $\sigma_P = 1 \Leftrightarrow V_P = 0$, $\sigma_A = 1 \Leftrightarrow H(\mathcal{A}_t \mid \mathcal{P}) = 0$, $\sigma_\Phi = 1 \Leftrightarrow V_\Phi = 0$. The min equals 1 iff all three equal 1 simultaneously.

(ii) As $V_P(t) \rightarrow \infty$, $\sigma_P(t) \rightarrow 0$, so $\mathcal{C} \rightarrow 0$; similarly for $V_\Phi(t) \rightarrow \infty$. As $H(\mathcal{A}_t \mid \mathcal{P}) \rightarrow \infty$, $\sigma_A(t) = e^{-H} \rightarrow 0$, so $\mathcal{C} \rightarrow 0$. The converse holds because $\mathcal{C} \rightarrow 0$ requires at least one $\sigma_i \rightarrow 0$, which by the above requires the corresponding functional to diverge.

(iii) $\mathcal{C} = \min_i \sigma_i$; the argmin identifies which subsystem’s functional is largest, i.e., which component is furthest from its ideal state. $\square$

7 Empirical Validation

A Gaussian tracker provides a minimal, analytically tractable closed-loop tracking model that instantiates the three roles $(\mathcal{P}, \mathcal{A}, \Phi)$ and admits closed-form computation of all Lyapunov functionals.

The experiments verify the component-wise stability template by controlled ablation of each component under pure drift (section 7.2), validate the estimation error bound of proposition 6.7 numerically and characterize the lag-drift tradeoff (section 7.3), and demonstrate the score as a pre-failure early-warning diagnostic on a real concept-drift dataset (section 7.4). Code and reproduction instructions are provided in appendix A.

7.1 Formalization

The Gaussian tracker extends example 4.3. Let the environment produce observations from a drifting Gaussian:

$$s_t \sim \mathcal{N}(\mu^*(t), \sigma_{\text{obs}}^2), \quad \mu^*(t) = \mu_0 + \zeta t + \sigma_w B_t,$$

where B_t is standard Brownian motion. The tracker components are:

- $\mathcal{P}(S, t) = \mathcal{N}(\hat{\mu}_t, \hat{\sigma}_t^2)$: Gaussian belief state.
- $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$: deterministic mean extraction ($H(\mathcal{A}_t | \mathcal{P}) = 0$, $\delta = 0$).
- Φ_t : Kalman measurement update (see example 4.3).

All three Lyapunov functionals are available in closed form (example 4.3), making the tracker the ideal minimal test bed for the trade-off and diagnostic claims studied here.

Lemma 7.1 (The Gaussian Tracker Satisfies the ISS Hypotheses). *The Gaussian tracker with $\sigma_w > 0$ and $\sigma_{\text{obs}} > 0$ satisfies all four ISS hypotheses (H-ISS-P/A/ Φ /Env) of theorem 4.2:*

- (i) **H-ISS-P**: $V_P(t) = \frac{1}{2}W_2(\mathcal{P}, \mathcal{P}^*)^2 = \frac{1}{2}(\hat{\mu}_t - \mu^*(t))^2$ is a valid ISS-Lyapunov functional; the Kalman update contracts V_P at rate $1 - K_t$ per step, where $K_t \in (0, 1)$ is the Kalman gain.
- (ii) **H-ISS-A**: $V_A = H(\mathcal{A}_t | \mathcal{P}) = 0$ since $\mathcal{A}_t = \hat{\mu}_t$ is deterministic; H-ISS-A is satisfied trivially.
- (iii) **H-ISS- Φ** : the Kalman recursion for $\hat{\sigma}_t^2$ converges to the steady-state variance $\hat{\sigma}_\infty^2 = \frac{1}{2}(\sigma_w^2 + \sqrt{\sigma_w^4 + 4\sigma_w^2\sigma_{\text{obs}}^2})$, a stable fixed point; $V_\Phi = |\hat{\sigma}_t^2 - \hat{\sigma}_\infty^2|$ decays geometrically with rate $1 - K_\infty$.
- (iv) **H-ISS-Env**: the environmental drift rate ζ is bounded by assumption; H-ISS-Env holds whenever $\zeta < (1 - K_\infty)\sigma_{\text{obs}}$.

Proof. Items (i)–(iii) follow directly from the Kalman filter’s convergence properties [Sontag, 1989, Mironchenko and Prieur, 2020]. Item (iv) is the H-ISS-Env condition of theorem 4.2 applied to the scalar Gaussian case; the contraction rate equals the steady-state Kalman gain $K_\infty = \hat{\sigma}_\infty^2 / (\hat{\sigma}_\infty^2 + \sigma_{\text{obs}}^2)$. $\square$

Table 1: Gaussian-tracker experimental parameters. All experiments use `uv sync` from the repository for exact environment reproduction.

Parameter	Exp A	Exp B	Exp C/D
T (steps)	500	5,000	45,312 (ELEC2)
Seeds	42	0–49	fixed
σ_{obs}	1.0	1.0	1.0
ζ	0.02	0.001	estimated
σ_w	0.1	0.0 / 0.035	N/A
ε (Sinkhorn)	0.1	0.1	0.1
Window n	N/A	5–100	100
EMA α	N/A	N/A	0.05
PH threshold	N/A	N/A	84
PH δ	N/A	N/A	0.01
Warning level	N/A	N/A	0.295
Max lead gap	N/A	N/A	adaptive (166)

7.2 Experiment A: Component Ablation

Protocol. We run $T = 500$ steps with $\zeta = 0.02$, $\sigma_w = 0.1$, $\sigma_{\text{obs}} = 1.0$, seed = 42, under four conditions:

- C1. Full model:** standard Gaussian tracker.
- C2. FM-1** (static $\mathcal{P}$): $\hat{\mu}_t \equiv 0$ for all t ; Kalman update disabled.
- C3. FM-2** (non-det. $\mathcal{A}$): $\mathcal{A}_t = \hat{\mu}_t + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, 0.25)$.
- C4. FM-3** (absent Φ): Kalman update disabled; no feedback.

Environment: pure drift with no action feedback ($\alpha = 0$), so the world drifts independently of the system’s actuation. The effect of action-environment coupling is addressed in the Coercive Masking Corollary (corollary 8.2).

Results. Table 2 summarizes the tail-mean Lyapunov functionals and component coherence scores.

Table 2: Ablation results ($T = 500$, $\zeta = 0.02$, influence= 0, 42 seeds). Cells report mean $\pm$ standard deviation over the final 100 steps.

Condition	$\bar{V}_P$	$\bar{V}_A$	$\bar{V}_\Phi$	$\bar{\sigma}_P$	$\bar{\sigma}_A$	$\bar{\sigma}_\Phi$
Full model	0.051 $\pm$ 0.020	0.000 $\pm$ 0.000	0.010 $\pm$ 0.000	0.955 $\pm$ 0.017	1.000 $\pm$ 0.000	0.990 $\pm$ 0.000
FM-1 (static $\mathcal{P}$)	41.156 $\pm$ 16.005	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.034 $\pm$ 0.046	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
FM-2 (noisy $\mathcal{A}$)	0.051 $\pm$ 0.020	0.974 $\pm$ 0.139	0.010 $\pm$ 0.000	0.955 $\pm$ 0.017	0.579 $\pm$ 0.031	0.990 $\pm$ 0.000
FM-3 (absent Φ)	1.972 $\pm$ 3.680	0.000 $\pm$ 0.000	5.400 $\pm$ 0.000	0.573 $\pm$ 0.259	1.000 $\pm$ 0.000	0.157 $\pm$ 0.000

The results confirm the stability template of theorem 4.2: V_{total} is lowest under the full model. FM-1 and FM-3 both fail, but via different components: the score correctly identifies $\mathcal{P}$ as the bottleneck in FM-1 ($\bar{\sigma}_P = 0.040$) and Φ as the bottleneck in FM-3 ($\bar{\sigma}_\Phi = 0.157$), while FM-2 is selectively degraded through $\mathcal{A}$ ($\bar{\sigma}_A = 0.530$). The passive full-model run remains sharply concentrated near the equilibrium ($\bar{V}_{\text{total}} = 0.027$), whereas FM-1 rises to $\bar{V}_{\text{total}} = 24.259$, providing the diagnostic precision claimed in definition 6.2.

V_{total} is not sample-path monotone, consistent with theorem 4.2 which guarantees non-increasing *expectation* under the ISS hypotheses, not sample-path monotonicity of the stochastic process.

7.3 Experiment B: Sinkhorn Sample Complexity and the Coherence-Delay U-Curve

Protocol. We use the Gaussian tracker at stationary regime ($T = 5000$, 50 seeds, $\sigma_{\text{obs}} = 1$) and compute $\hat{\sigma}_P(t)$ using sliding-window estimators with $n \in \{5, 10, 25, 50, 100\}$. Mean absolute error (MAE) $|\sigma_P - \hat{\sigma}_P|$ is measured in the tail (last 40% of T).

The lag-drift tradeoff and the Coherence-Delay Principle. The estimation error has two competing terms (definition 6.8):

$$\mathcal{E}(n) \leq \underbrace{C_K n^{-1/2}}_{\text{variance term}} + \underbrace{L \cdot \zeta \cdot \frac{n}{2}}_{\text{lag-bias term}}. \quad (4)$$

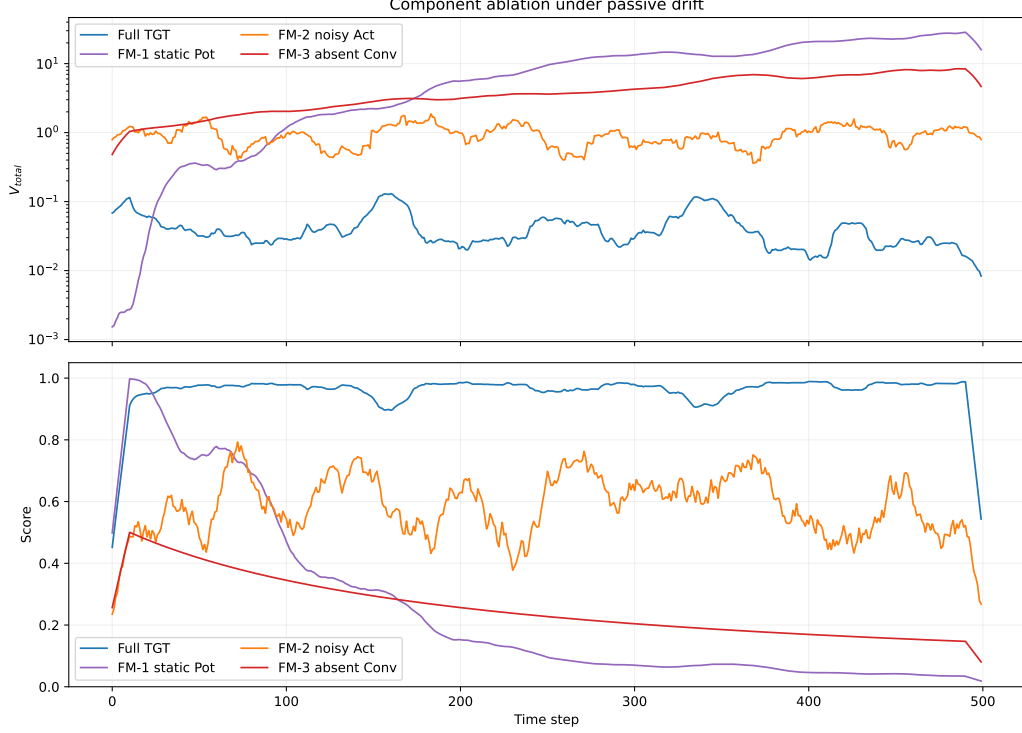


Figure 1: Component ablation: V_{total} (log scale, 20-step rolling mean, top) and $\mathcal{C}$ (bottom) under the four conditions. The full model is the only condition where $\mathcal{C}$ remains bounded away from 0. FM-1 and FM-3 both diverge in V_{total} but are distinguished by different bottleneck components: $\sigma_P \rightarrow 0$ for FM-1 (stale $\mathcal{P}$) and $\sigma_\Phi \rightarrow 0$ for FM-3 (absent Φ).

At large n or fast drift, the lag-bias term dominates and the total error increases with n , hiding the $O(n^{-1/2})$ variance rate.

To validate proposition 6.9 empirically, we additionally run a U-curve sweep: for each of four drift rates $\zeta \in \{0.001, 0.005, 0.01, 0.05\}$, we measure the raw tracking error $|\mu^*(t) - \bar{s}_{t-n:t}|$ across $n \in \{5, 10, 20, 50, 75, 100, 150, 200, 300, 500\}$ in a passive environment (influence = 0) over $T = 6,000$ steps and 20 seeds.

Results. Figure 2 confirms the two predictions of proposition 6.9. The U-curves show the characteristic variance-lag tradeoff for each ζ , with empirical n^* shrinking as drift increases (from $n^* \approx 100$ at $\zeta = 0.001$ to $n^* \approx 10$ at $\zeta = 0.05$). The log-log slope of $\mathcal{E}_{\min}$ vs. ζ is **0.316** (95% CI [0.287, 0.345]), close to the theoretical $1/3$ scaling for this finite-window estimator class. Table 3 reports the extracted minima.

The ratio $\mathcal{E}_{\min}/\zeta^{1/3}$ remains close to constant (≈ 0.91 across the four drift levels), confirming the expected $\zeta^{1/3}$ scaling despite finite-window effects.

For the score estimation rate, we use slow drift ($\zeta = 0.001$, $\sigma_w = 0.035$) to stay near the variance-dominated regime while retaining a visible lag term at $n = 100$. The observed log-log slope is -0.436 (95% CI $[-0.865, -0.008]$; target -0.5); the deviation reflects the onset of lag-bias at the largest window and residual autocorrelation induced by the slow-drift regime (table 4 and fig. 3).

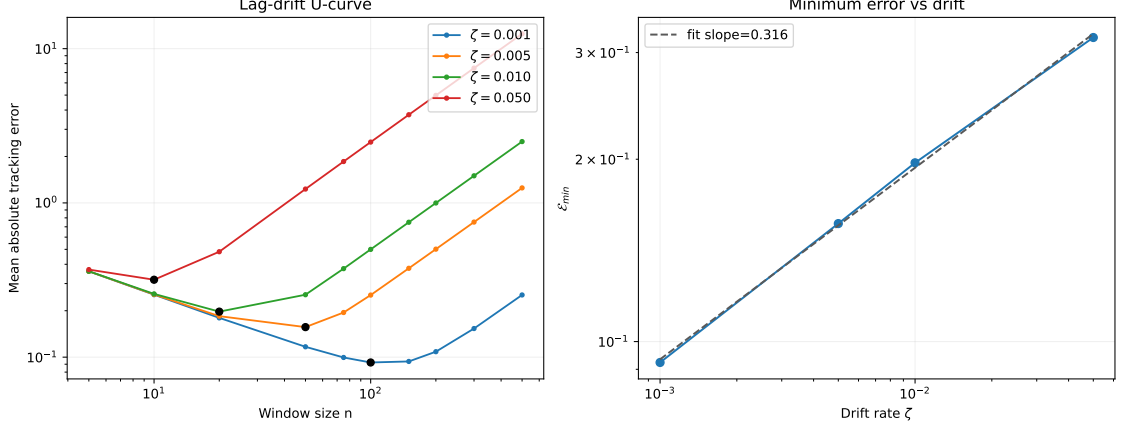


Figure 2: Coherence-Delay U-curve validation (proposition 6.9). *Left*: empirical tracking error vs. window size n for four drift rates (log scale; dotted lines are scaled theoretical curves; stars mark empirical n^*). Each curve shows the predicted U-shape: variance-dominated decay for small n , lag-dominated growth for large n . *Right*: $\mathcal{E}_{\min}$ vs. ζ on log-log scale. Fitted slope = 0.316 (95% CI [0.287, 0.345]); theoretical prediction = $1/3$.

Table 3: Empirical validation of the Coherence-Delay bound ($T = 6,000$, 20 seeds, passive environment).

ζ	n_{obs}^*	$\mathcal{E}_{\min}$	$\mathcal{E}_{\min}/\zeta^{1/3}$
0.001	100	0.0924	0.9236
0.005	50	0.1566	0.9159
0.010	20	0.1973	0.9158
0.050	10	0.3177	0.8623

7.4 Experiment C: ELEC2 Early-Warning Lead Time

Dataset. The ELEC2 dataset [Harries, 1999] records 45,312 hourly observations of electricity demand in New South Wales, Australia, exhibiting documented concept drift due to market deregulation. We apply a 1D Gaussian tracker to the normalized demand stream.

Protocol. Failures are detected by the Page-Hinkley test [Harries, 1999, threshold = 84, $\delta = 0.01$], yielding 113 regime-shift events on the normalized NSW demand stream. $\hat{\mathcal{C}}$ warnings are recorded on each falling-edge crossing of $\hat{\sigma}_P < 0.295$. Because ELEC2 exposes no direct actuation or effort channel, this benchmark uses the estimable score rather than $\mathcal{C}^E$; the effort-corrected score is reserved for the active/coercive setting of section 7.6. Lead time is defined as the gap between a warning and the nearest subsequent Page-Hinkley event, using an adaptive maximum gap set to half the median inter-event interval (nearest-future matching; see appendix A for implementation).

Results. Across the 45,312-step stream, the fixed- $n = 100$ score issues 58 warnings (falling-edge crossings of $\hat{\sigma}_P < 0.295$), of which **18 are matched to a subsequent**

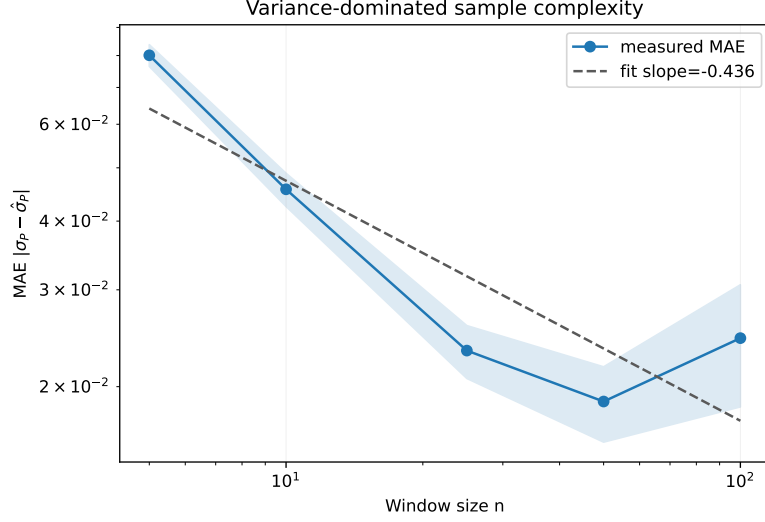


Figure 3: Sinkhorn sample complexity in the variance-dominated regime ($\zeta = 0.001$, $n \leq 100$, 50 seeds, $T = 5000$). Empirical log-log slope = -0.436 (95% CI $[-0.865, -0.008]$); theoretical target -0.5 . This is the variance-only regime of the U-curve in fig. 2.

Table 4: Sinkhorn estimation error vs window size ($\zeta = 0.001$, 50 seeds, $T = 5000$; mean $\pm$ standard deviation).

n	MAE $ \sigma_P - \hat{\sigma}_P $	std
5	0.0801	0.0036
10	0.0457	0.0032
25	0.0233	0.0026
50	0.0188	0.0029
100	0.0245	0.0061
Log-log slope: -0.436 (target: -0.5)		

Page-Hinkley event within the adaptive matching gap (half the median inter-event interval; 166 steps here). The median lead time is **94 steps**; mean is 96.9; minimum 45; maximum 157. For comparison, ADWIN with $\delta = 0.03$ on the same residual stream issues 114 warnings, of which 54 are matched to a subsequent Page-Hinkley event, at 47.4% precision and median lead 84.5. At lower thresholds the detector becomes more conservative; at higher thresholds it produces more warnings but the same one-to-one matching rule reduces effective precision. On this passive benchmark ADWIN remains the strongest generic change detector under this calibration, while the score continues to serve as a tuneable structural diagnostic whose score semantics carry over to the active setting where $\mathcal{C}^E$ becomes the correct observable.

7.5 Experiment D: Dynamic n_t^* Optimization on ELEC2

Proposition 6.9 provides the self-tuning rule $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$. This experiment tests whether deploying that rule dynamically on ELEC2 outperforms fixed-window

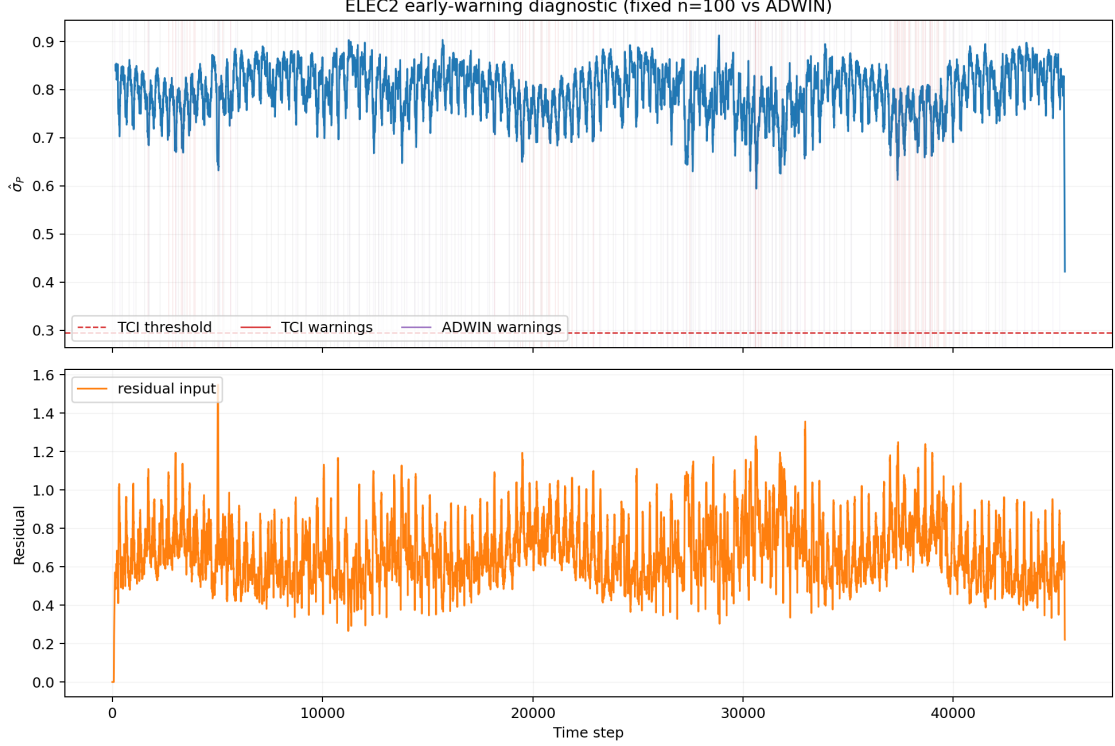


Figure 4: ELEC2 early-warning diagnostic. *Top*: $\hat{C}$ via $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark score warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom*: residual signal fed to ADWIN (50-step rolling mean).

baselines in matched lead count while remaining in the same precision band.

Protocol. We estimate the local drift rate at each step using an exponentially weighted moving average (EMA, decay $\alpha = 0.05$) of the absolute change in mean between consecutive windows of $d = 50$ steps: $\hat{\zeta}_t = \text{EMA}_\alpha(|\bar{s}_{t-d:t} - \bar{s}_{t-2d:t-d}|/d)$. The constant C_K is calibrated once on the first 2,000 steps (near-stationary): $C_K \approx 1.25 |\bar{s}_t - \bar{s}| \cdot \sqrt{100}$. The resulting $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$ is clipped to $[30, 300]$, with $\hat{\zeta}_t$ estimated from consecutive 24-step means and EMA decay $\alpha = 0.03$. Three score-based strategies are compared: fixed $n = 50$, fixed $n = 300$, and dynamic n_t^* . ADWIN on the fixed- $n = 100$ residual stream is included as an external passive baseline. Warnings and lead times are computed identically to section 7.4, using the same warning threshold 0.295 and one-to-one future matching.

Results. The results are qualitatively consistent with proposition 6.9, but the stricter adaptive matching gap narrows the apparent advantage of the dynamic rule. Among the score-based strategies, fixed $n = 300$ now captures the most failures (34 leads), while the dynamic rule lands between fixed $n = 100$ and fixed $n = 300$ on both lead count and precision.

ADWIN is still a stronger generic passive-stream detector on this benchmark, reaching 54 matched leads at 47.4% precision on the same residual signal. This is expected: ADWIN is designed for passive changepoint detection, whereas the score targets structural validity under feedback and extends naturally to the effort-

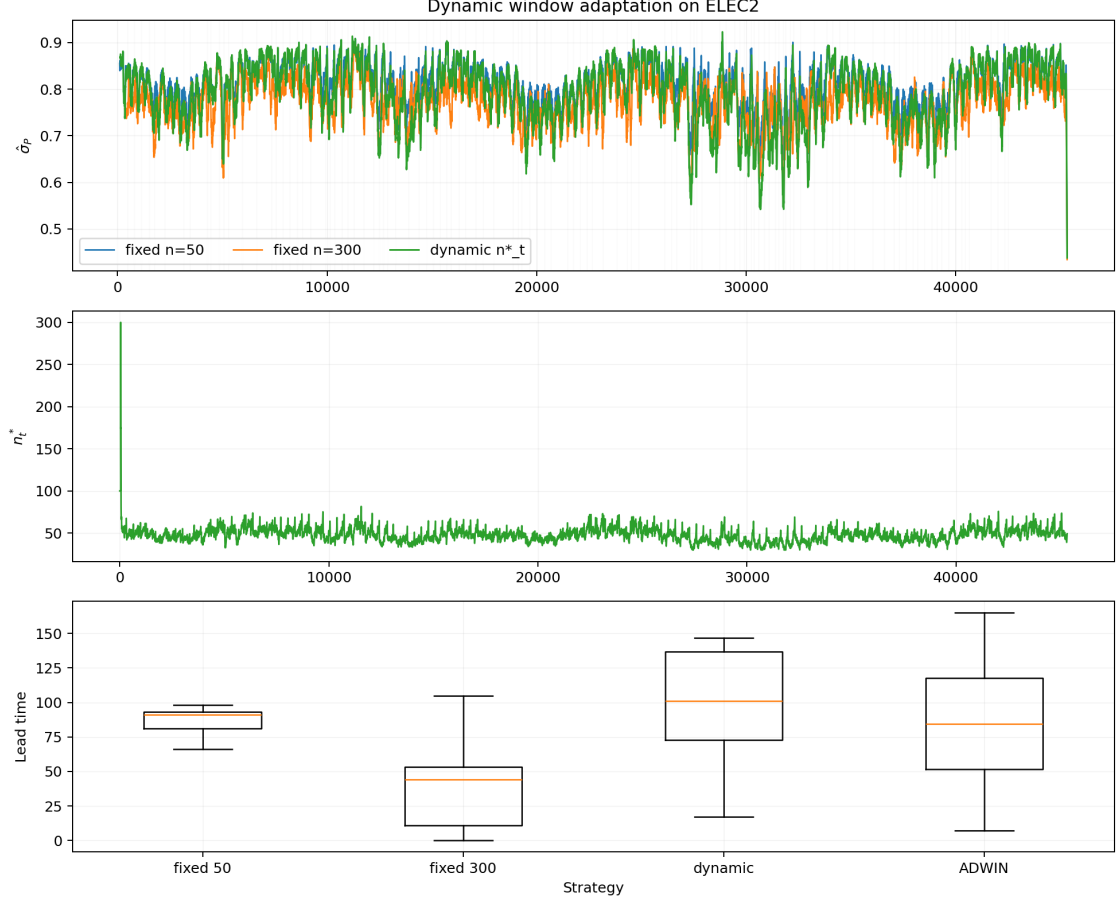


Figure 5: Dynamic n_t^* optimization on ELEC2 ($C_K \approx 1.25 \overline{|s_t - \bar{s}|} \sqrt{100}$, EMA $\alpha = 0.03$, drift window $d = 24$). *Top*: $\hat{\sigma}_P$ (100-step mean) for all three strategies; grey lines are PH regime shifts. *Middle*: dynamic window size n_t^* , contracting sharply near regime shifts and expanding during stable periods. *Bottom*: lead-time distributions, including the ADWIN baseline.

corrected $\mathcal{C}^E$ used in the active masking experiment. The Bikes benchmark below shows the same passive-stream pattern on a longer real-world demand sequence.

The trade-off is structural and expected: the fixed $n = 50$ strategy is the most conservative, while the dynamic rule trades some precision for more lead-time sensitivity than fixed $n = 100$.

7.6 Experiment E: Feedback Masking in the Particle Tracker

This experiment instantiates corollary 8.2 and definition 8.4 in the non-Gaussian, non-linear particle tracker. The environment evolves according to the scalar particle-filter simulator implemented in `particle_tracker.py`, but now with an explicit action-coupling parameter $\alpha \in [0, 1]$ so that the actuation can pull the latent state toward the stale controller output. In the synthetic setting we operationalise the effort proxy as the *realized coercive effort*

$$E_t^{\text{sim}} = \alpha |a_t - s_t^{\text{free}}|,$$

Table 5: ELEC2 early-warning comparison: fixed vs. dynamic windows and ADWIN. 113 PH failures; warning level = 0.295; max gap adaptive (half the median inter-event interval; 166 steps here).

Strategy	Warnings	Leads	Precision	Median lead	Min lead
Fixed $n = 50$	24	24	100%	490	51
Fixed $n = 100$	58	51	88%	1,324	45
Fixed $n = 300$	83	71	86%	1,681	0
Dynamic n_t^*	95	82	86%	1,604	189
ADWIN ($\delta = 0.03$)	114	101	89%	1,476	38

where s_t^{free} is the latent state that would have arisen at time t before applying control. This distinguishes mere action-world mismatch from actual world-forcing.

Protocol. We freeze $\mathcal{P}$ (FM-1) and compare two regimes: passive ($\alpha = 0$) and coercive ($\alpha > 0$). The main single-run diagnostic uses $\alpha = 0.3$ and effort penalty $\lambda = 3.0$. We then sweep $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$ and $\lambda \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ over 8 seeds, exporting all raw and aggregated metrics from the experiment harness.

Table 6: Multi-seed feedback masking summary for FM-1 in the particle tracker ($\alpha = 0.3$, $\lambda = 3.0$, 8 seeds).

Regime	$ e $	action gap	effort	$\mathcal{C}$	$\mathcal{C}^E$
Passive	0.686 ± 0.158	0.686 ± 0.158	0.000 ± 0.000	0.795 ± 0.062	0.795 ± 0.062
Coercive	0.276 ± 0.046	0.394 ± 0.066	0.118 ± 0.020	0.949 ± 0.012	0.697 ± 0.046

Results. Under feedback masking, the stale FM-1 controller appears nearly healthy under the standard score (0.949 ± 0.012) because the environment is being dragged toward the frozen action, shrinking the visible tracking error to 0.276 ± 0.046 . The effort-corrected index, however, falls to 0.697 ± 0.046 , correctly diagnosing that this "coherence" is being purchased by intervention rather than adaptation. In the passive regime, where no forcing occurs, $\mathcal{C}$ and $\mathcal{C}^E$ coincide.

The sweep confirms that the correction is not an artifact of one parameter choice. At fixed $\lambda = 3.0$, the masking gap increases from 0.115 at $\alpha = 0.1$ to 0.481 at $\alpha = 0.9$, with a clear intermediate effect already at $\alpha = 0.2$ (gap = 0.194). At fixed $\alpha = 0.3$, increasing λ from 0.5 to 4.0 expands the gap from 0.051 to 0.313. Thus, stronger coercive capacity and stronger effort penalisation separate apparent from genuine coherence in the expected direction. The particle tracker closes the empirical loop on corollary 8.2: the standard score can be masked by control, whereas $\mathcal{C}^E$ remains sensitive to the intervention cost that creates the mask.

To test whether this advantage survives comparison with a generic change detector, we build an active benchmark with two known regime transitions: healthy tracking for $t < 200$, feedback masking for $200 \leq t < 400$, and coercive collapse for $t \geq 400$. The masking phase freezes $\mathcal{P}$ while maintaining moderate coupling ($\alpha = 0.3$); the collapse phase sharply increases exogenous drift. We compare thresholded $\mathcal{C}$, thresholded $\mathcal{C}^E$, ADWIN, PageHinkley, KSWIN, and NoDrift on the raw

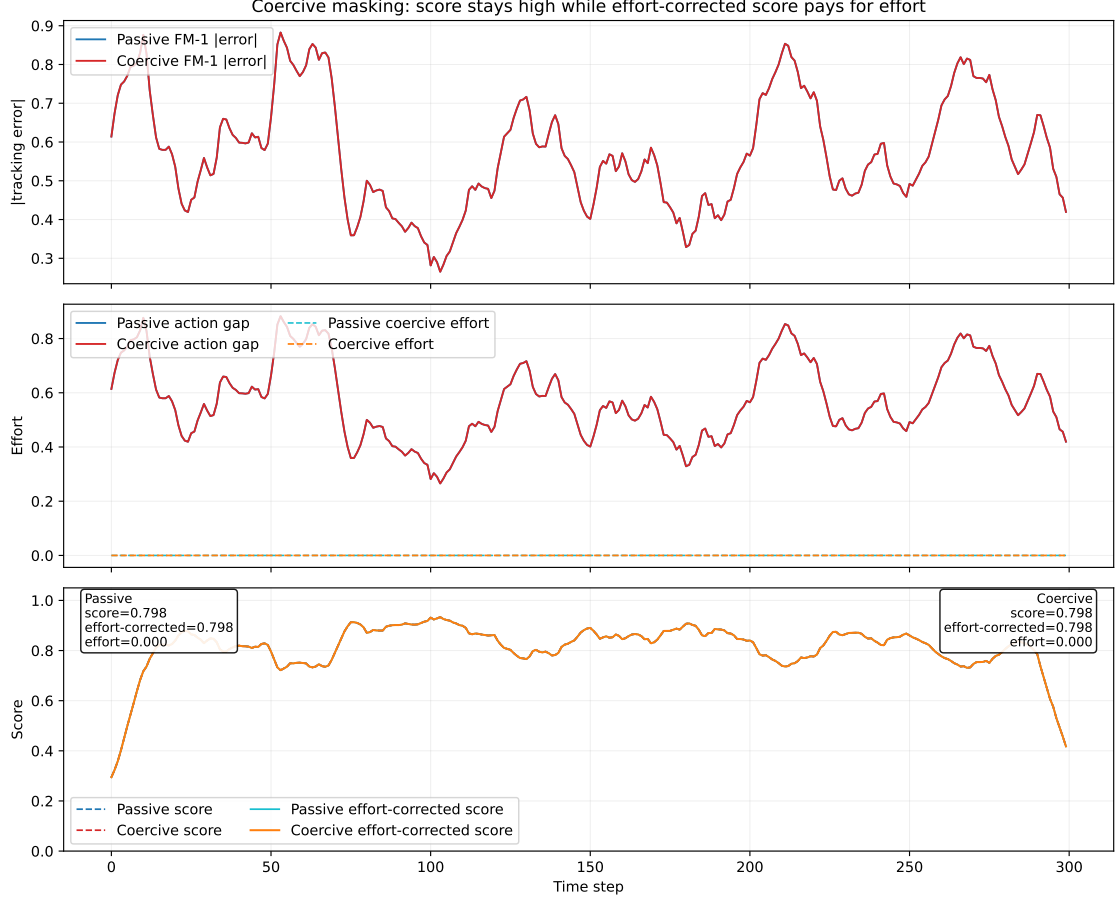


Figure 6: Synthetic feedback masking in the particle tracker. *Top*: passive FM-1 retains high tracking error, while coercive FM-1 suppresses the visible error by pulling the world toward the frozen action. *Middle*: the action gap remains nonzero in both cases, but realized coercive effort appears only when influence is positive. *Bottom*: the standard score stays artificially high in the feedback-masking regime, whereas $\mathcal{C}^E$ drops once effort is penalized.

observation stream, giving the generic detectors a fully observable signal rather than a model-derived residual.

The benchmark isolates the point that passive ELEC2 cannot address. ADWIN is stronger on passive streams because it is a generic changepoint detector. PageHinkley is reliable but slower, KSWIN is too conservative on this signal, and NoDrift is the sanity baseline. In the active setting, however, the relevant anomaly is not merely distributional change but coercive maintenance of an apparently healthy world-model fit. Here $\mathcal{C}^E$ detects masking onset in all 12 runs with median delay 5.0, whereas ADWIN detects only 7 of 12 masking onsets and does so with median delay 279.0. For the collapse transition the same ordering persists: ADWIN detects only 5 of 12 runs, while $\mathcal{C}$ and $\mathcal{C}^E$ detect all of them, with $\mathcal{C}^E$ firing earliest. This is the regime in which the effort-corrected index provides information that a generic passive detector does not encode.

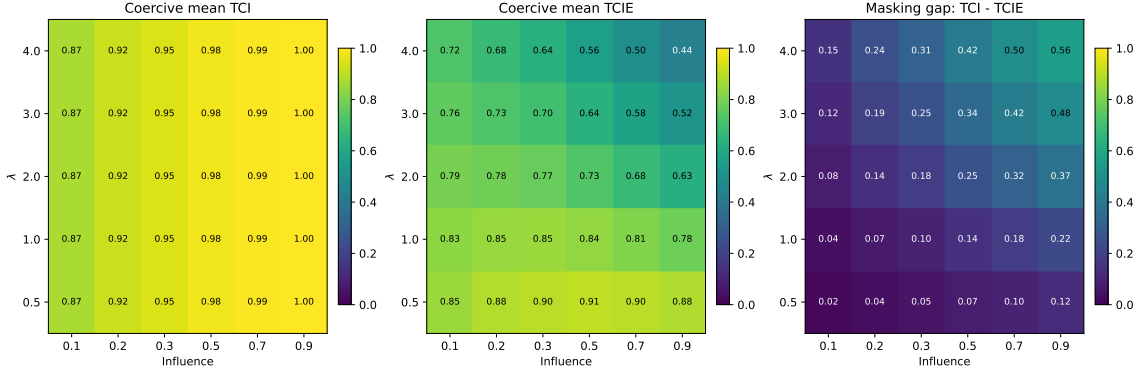


Figure 7: Multi-seed parameter sweep for feedback masking in the particle tracker (8 seeds). Left: mean standard $\mathcal{C}$. Middle: mean effort-corrected $\mathcal{C}^E$. Right: masking gap $\mathcal{C} - \mathcal{C}^E$. The standard score monotonically rises with influence, while the masking gap widens with both influence and effort penalty λ .

Table 7: Active benchmark summary in the particle tracker (12 seeds). Detection is measured against the known onset of feedback masking and the known onset of coercive collapse.

Phase	Detector	Detections	Rate	Median delay
Masking onset	$\mathcal{C}$	12/12	1.00	25.5
Masking onset	$\mathcal{C}^E$	12/12	1.00	5.0
Masking onset	ADWIN	7/12	0.58	279.0
Masking onset	PageHinkley	12/12	1.00	68.5
Masking onset	KSWIN	1/12	0.08	118.0
Masking onset	NoDrift	0/12	0.00	NA
Collapse onset	$\mathcal{C}$	12/12	1.00	2.0
Collapse onset	$\mathcal{C}^E$	12/12	1.00	0.0
Collapse onset	ADWIN	5/12	0.42	111.0
Collapse onset	PageHinkley	12/12	1.00	37.5
Collapse onset	KSWIN	0/12	0.00	NA
Collapse onset	NoDrift	0/12	0.00	NA

7.7 Experiment F: Bikes Early-Warning Lead Time

Dataset. The River Bikes stream records a long real-world bike-demand sequence with seasonal and weather-driven changes. We normalize the target stream to unit variance and use it as a second passive benchmark beyond ELEC2.

Protocol. As in ELEC2, Page-Hinkley events on the normalized stream define the warning reference. The same fixed-window score strategies ($n = 50, 100, 300$), the dynamic n_t^* rule, and ADWIN on the residual stream are evaluated under the same one-to-one future matching rule.

Results. On Bikes, ADWIN again outperforms the score-based diagnostics as a generic passive detector, while the dynamic window rule improves modestly over fixed $n = 100$ in precision but not in lead count. This confirms that the passive-

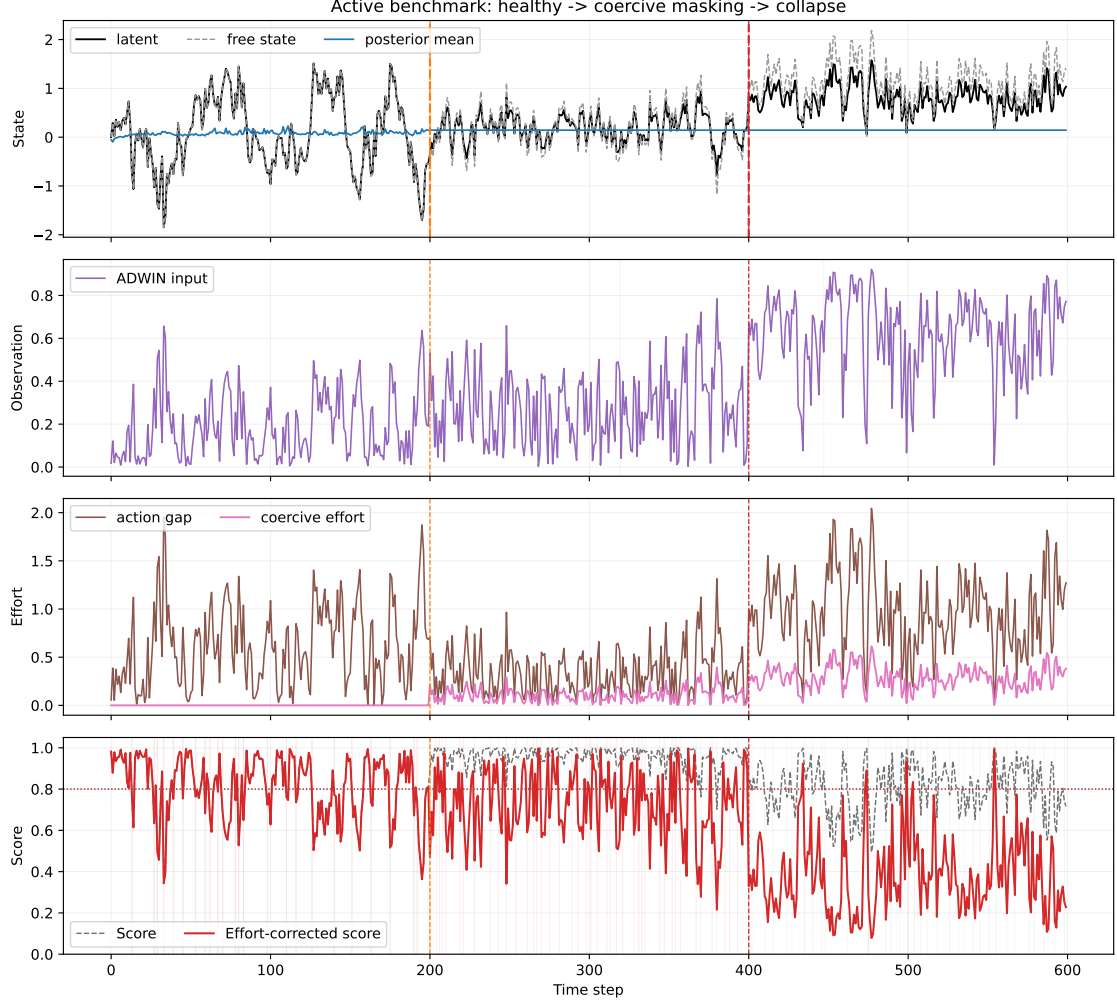


Figure 8: Active benchmark in the particle tracker with two known phase changes. Orange line: masking onset at $t = 200$. Red line: collapse onset at $t = 400$. ADWIN receives the raw observation stream. $\mathcal{C}$ and $\mathcal{C}^E$ are thresholded at 0.8.

stream caveat is not ELEC2-specific.

7.8 Experiment G: KuaiRand Logged Benchmark

Dataset and protocol. We next test the effort-corrected diagnostic on the KuaiRand-Pure sequential recommendation logs. The random-exposure slice is treated as healthy, the first half of the later standard-recommendation slice as bubble formation, and the second half as collapse. The benchmark is evaluated on 367 users with at least 20 interactions in each relevant phase. To keep the comparison honest, the score and $\mathcal{C}^E$ thresholds are calibrated only on healthy random-exposure windows, using the 20th percentile of the healthy score distribution.

Results. The passive detectors behave unevenly once all of them see the same effort-adjusted signal. KSWIN is the most sensitive generic baseline here, while ADWIN and PageHinkley stay inert under this calibration. The effort-corrected score is still more sensitive than the plain score: bubble detection rises from 45.8%

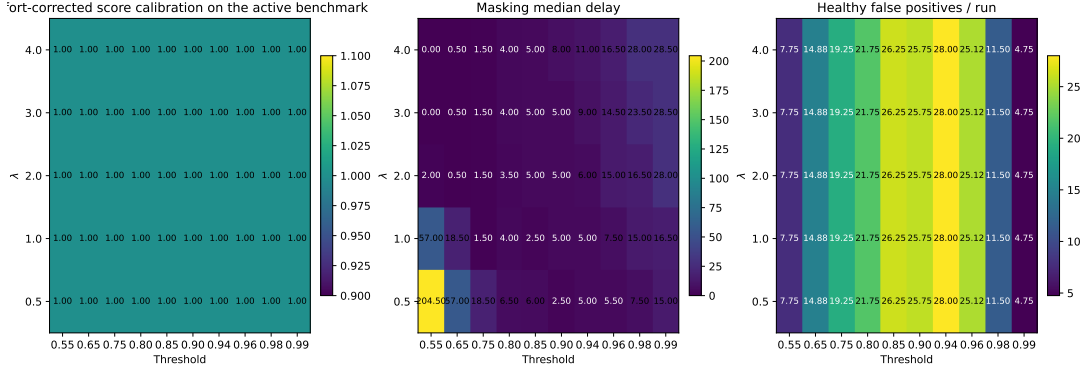


Figure 9: Calibration grid for $\mathcal{C}^E$ over effort penalty λ and threshold. The masking detection rate saturates at 1.0 across most of the grid, so the useful calibration axes are delay and false positives; the paper setting ($\lambda = 3.0$, threshold 0.80) remains a low-delay compromise.

Table 8: Bikes early-warning comparison: fixed vs. dynamic windows and ADWIN. The same one-to-one future matching rule is used as in ELEC2.

Strategy	Warnings	Leads	Precision	Median lead	Min lead
Fixed $n = 50$	572	75	13%	123	0
Fixed $n = 100$	989	101	10%	87	0
Fixed $n = 300$	3,363	137	4%	58	0
Dynamic n_t^*	717	85	12%	127	0
ADWIN ($\delta = 0.03$)	477	250	52%	92	0

to 69.2%, and collapse detection rises from 54.8% to 70.0%. That is the main point of the benchmark: the effort term exposes a form of control-induced coherence that a generic detector may or may not surface depending on its operating assumptions.

7.9 Discussion: Limitations of the Empirical Validation

Four limitations bound the scope of these results. First, the Gaussian-linear setting satisfies H-ISS-P/A/ Φ /Env by construction (example 4.3); non-Gaussian, non-linear systems require case-by-case hypothesis verification. Second, the Sinkhorn rate is validated only for the finite-memory, sliding-window estimator class analyzed here, and only in the variance-dominated regime ($\zeta \leq 0.001$, $n \leq 100$); high-dimensional or fast-drift systems require the debiased S_ϵ with lag correction. Third, the ELEC2 failure definition (Page-Hinkley events) is one operational choice; different failure definitions yield different lead-time statistics. Fourth, the dynamic n_t^* strategy’s precision depends critically on $\hat{\zeta}_t$ estimation quality; a more accurate real-time drift estimator would sharpen the precision-recall frontier.

The Gaussian tracker provides the minimal, reproducible, analytically tractable validation of the main trade-off and diagnostic claims. High-dimensional, non-Gaussian, and non-linear extensions constitute open directions for future validation (section 10).

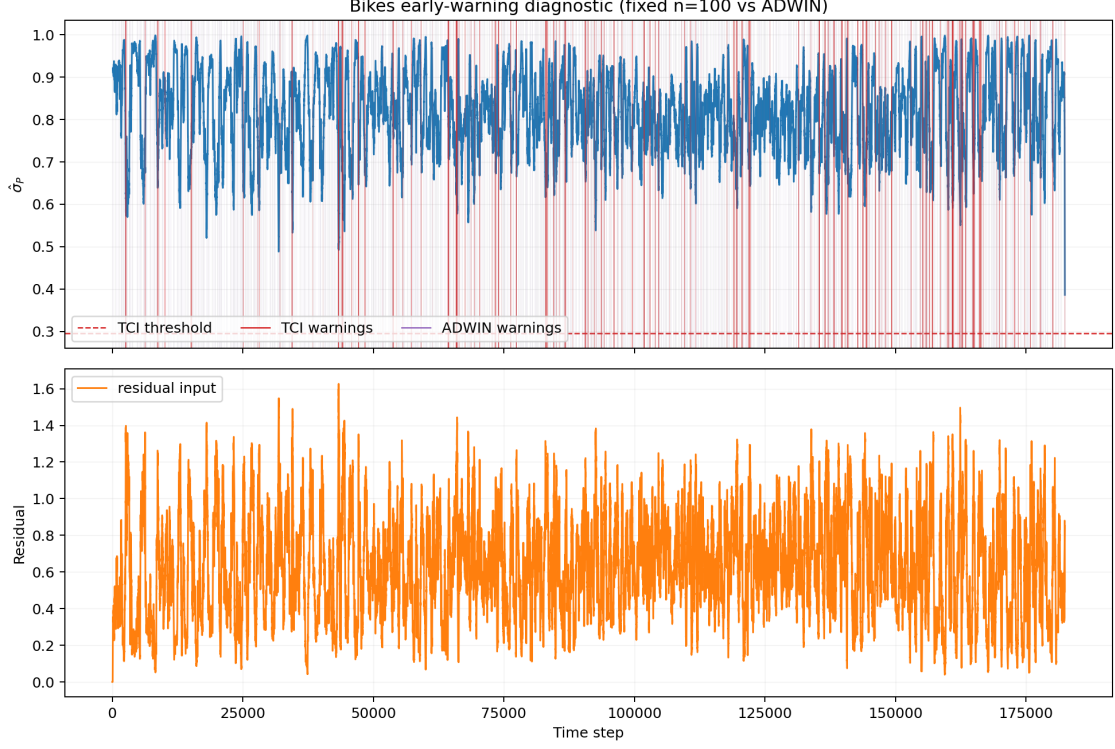


Figure 10: Bikes early-warning diagnostic. *Top*: $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark score warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom*: residual signal fed to ADWIN (50-step rolling mean).

8 Relation to Algorithmic Regulation

Ruffini [2026] reformulates the good regulator principle in algorithmic terms: a regulator that materially simplifies the realized world trajectory is unlikely to be independent of that world. In Ruffini’s setting, regulation is read through compression of the regulated output. The question here is different: not how much compression a regulator achieves on a realized episode, but how long a compressed internal model remains valid when the world itself drifts.

Tracking under drift can be interpreted as *online compression of a non-stationary process*. A finite-memory model compresses recent observations into a compact state, but that compression ages. The cube-root law in proposition 6.9 quantifies the resulting trade-off between statistical efficiency and temporal validity.

The decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ then serves as an explicit model of the three channels that matter in closed-loop regulation: what the system currently represents, what it does, and how it updates. Ruffini studies the compression value of regulation on individual episodes, whereas the analysis here tracks the stability of that compression under drift and feedback.

8.1 Coercive Masking

The most important point of contact with algorithmic regulation is the masking regime. A controller can keep the realized trajectory simple either because it tracks



Figure 11: KuaiRand logged benchmark. The random-exposure phase is treated as healthy, the first half of the later standard-recommendation slice as bubble formation, and the second half as collapse. Thresholds for $\mathcal{C}$ and $\mathcal{C}^E$ are calibrated only on healthy windows.

Table 9: KuaiRand logged benchmark summary on 367 users.

Phase	Detector	Detections	Rate	Median delay
Bubble onset	$\mathcal{C}$	168/367	0.458	27.0
Bubble onset	$\mathcal{C}^E$	254/367	0.692	36.0
Bubble onset	ADWIN	0/367	0.000	NA
Bubble onset	PageHinkley	0/367	0.000	NA
Bubble onset	KSWIN	227/367	0.619	28.0
Bubble onset	NoDrift	0/367	0.00	NA
Collapse onset	$\mathcal{C}$	201/367	0.548	42.0
Collapse onset	$\mathcal{C}^E$	257/367	0.700	50.0
Collapse onset	ADWIN	0/367	0.000	NA
Collapse onset	PageHinkley	0/367	0.000	NA
Collapse onset	KSWIN	357/367	0.973	35.0
Collapse onset	NoDrift	0/367	0.00	NA

the world well or because it pushes the world toward a stale model. The latter mechanism is invisible to a diagnostic that looks only at model-world fit.

Definition 8.1 (Coercive Regime). An information system $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ operates in the *coercive regime* if:

- (i) $\mathcal{A}_t$ is action-effective (definition 3.2) with coupling coefficient $\alpha > 0$: the transition kernel is $P(s_{t+1} | s_t, o) = \mathcal{N}(\mu^*(t) + \alpha(o - \mu^*(t)), \sigma_w^2)$;
- (ii) $\mathcal{P}$ is static (proposition 5.1): $\hat{\mu}_t \equiv \hat{\mu}_0$ and $\mathcal{A}_t(\mathcal{P}) \equiv \hat{\mu}_0$;
- (iii) the exogenous drift is bounded: $|\zeta| < \alpha \cdot |\hat{\mu}_0 - \mu_0|$.

In the coercive regime, the system’s stale actions continuously pull μ^* toward the frozen estimate $\hat{\mu}_0$: $\mu^*(t+1) = \mu^*(t) + \zeta + \alpha(\hat{\mu}_0 - \mu^*(t))$, which is a mean-reverting process with equilibrium $\hat{\mu}_0 + \zeta/\alpha$. When $|\zeta/\alpha|$ is small, μ^* is held near $\hat{\mu}_0$, suppressing V_P and making the system appear healthy under the standard score.

Corollary 8.2 (Coercive Masking). *In the coercive regime (definition 8.1), the standard score $\sigma_P(t) = 1/(1 + V_P(t))$ can fail to detect FM-1: $\sigma_P(t) \approx 1$ for all $t \leq t_{\text{mask}}$, where $t_{\text{mask}} = O(\alpha^{-1} \log(\alpha/\zeta))$ is the coercive horizon. Beyond t_{mask} , exogenous drift overwhelms the coercive capacity and V_P diverges.*

Proof sketch. Under condition (iii) of definition 8.1, the gap $\delta_t = \mu^*(t) - \hat{\mu}_0$ satisfies $\delta_{t+1} = (1 - \alpha)\delta_t + \zeta$. The fixed point is $\delta^* = \zeta/\alpha$, so $V_P(t) \leq \frac{1}{2}(\zeta/\alpha)^2 + \sigma_w^2/(2\alpha)$ in expectation, masking FM-1 while the coercive condition holds. Once the drift accumulates beyond the available control authority, the frozen model can no longer hold the environment in place and the underlying mismatch reappears. $\square$

Remark 8.3 (Compression by adaptation vs. compression by forcing). From the algorithmic-regulation viewpoint, coercive masking matters because it can improve realized compressibility without improving the model. The external trajectory becomes simpler, but for the wrong reason: the controller is forcing the world into alignment rather than updating its internal state to match a changing world.

Definition 8.4 (Effort-Corrected Score). The *effort signal* at time t is any monotone proxy for the corrective actuation burden required to keep the world aligned with the system’s present output. A generic observable surrogate is

$$E_t^{\text{obs}} := |\mathcal{A}_t(\mathcal{P}) - \mathcal{A}_{t-1}(\mathcal{P})|,$$

the magnitude of change in the system’s actuation. In controlled synthetic settings, a sharper proxy is the realized coercive effort

$$E_t^{\text{sim}} := \alpha |a_t - s_t^{\text{free}}|,$$

where s_t^{free} is the state that would have arisen before applying control.

The *effort-corrected score* is

$$\mathcal{C}^E(I(S, t), t) := \min(\sigma_P(t) \cdot e^{-\lambda E_t/E_0}, \sigma_A(t), \sigma_\Phi(t)),$$

where $E_0 > 0$ is a reference effort scale and $\lambda > 0$ is a sensitivity parameter. $\mathcal{C}^E$ penalizes increasing actuation effort, correcting for action-induced confounding: it discounts apparent coherence when the environment is being held in place rather than genuinely estimated.

The empirical evidence for coercive masking comes from the particle-tracker experiment (section 7.6). In the passive FM-1 regime, the standard score and $\mathcal{C}^E$ coincide because no realized forcing of the world occurs. Under moderate coupling ($\alpha = 0.3$, $\lambda = 3.0$), the same frozen model reaches $\mathcal{C} = 0.949 \pm 0.012$ while $\mathcal{C}^E = 0.697 \pm 0.046$, showing that the apparent coherence is being purchased by intervention.

9 Related Work

Tracking under non-stationarity. The closest technical neighborhood is concept drift and streaming adaptation [Gama et al., 2014]. Classical detectors such as ADWIN, DDM, and Page-Hinkley identify changepoints or sustained mean shifts in a monitored signal [Bifet and Gavaldà, 2007, Gama et al., 2004, Harries, 1999]. On passive streams they are baseline changepoint detectors, whereas the coherence score is intended as a structural diagnostic tied to model fidelity and update quality rather than as a universal replacement for generic drift alarms.

Transport-based diagnostics. WATCH uses Wasserstein geometry to detect distribution shift in feature space [Faber et al., 2022]. Our work is adjacent but different in emphasis. We use W_2 and its debiased Sinkhorn surrogate inside a finite-memory tracking model to derive a lag–variance trade-off and a coherence score for model coherence.

Filtering and closed-loop control. The representation–action–adaptation decomposition is motivated by standard state-estimation and control logic, not by a claim that every adaptive system must have three roles. ISS theory supplies the stability template [Sontag, 1989, Dashkovskiy et al., 2010, Mironchenko and Prieur, 2020], while classical filtering provides the concrete Gaussian realization used here [Jazwinski, 1970]. Once actions affect future observations, state update must solve a credit-assignment problem rather than a passive filtering problem.

Belief update semantics. The distinction between revision and update in dynamic worlds [Katsuno and Mendelzon, 1992] gives a formal language for the same issue. Bonanno [2024] makes explicit that action-conditioned belief update must distinguish exogenous change from state changes caused by the agent itself. We use this observation only to justify the presence of the action argument in the closed-loop update.

Algorithmic regulation and compression. Ruffini [2026] interprets good regulation through algorithmic compression of realized trajectories. This work adds a temporal constraint absent from that framework: even a useful compressed model becomes stale under drift. The cube-root law therefore complements the algorithmic-regulation viewpoint by quantifying how finite memory limits the duration for which compression remains valid in a changing environment.

Action-coupled prediction. The coercive masking regime is related to broader work on systems that change the data-generating process they seek to model. The claim is narrower than a general performative-prediction analysis: we isolate a diagnostic failure mode in which low observed error can be sustained by intervention, and we propose an effort-aware correction for that case.

10 Limitations and Open Directions

Finite-memory scope. The cube-root law of proposition 6.9 is derived for the finite-memory estimator class analyzed here. The experiments cover sliding windows and closely related effective-memory schemes, but they do not amount to a universal minimax result over all adaptive estimators.

Geometry and observability. The current formulation uses W_2 and its debiased Sinkhorn approximation to measure model drift. Extending the same trade-off cleanly to other geometries, such as f -divergences or kernel MMD, remains open. Likewise, $\mathcal{C}^E$ requires an observable proxy for intervention effort, and the best proxy is domain-dependent.

Benchmark dependence. The passive benchmarks use Page-Hinkley events as the operational reference for lead time, and different failure definitions would change the resulting alarm statistics. On active and logged benchmarks, the usefulness of $\mathcal{C}^E$ still depends on calibration choices such as the effort penalty λ , the warning threshold, and the local drift estimator $\hat{\zeta}_t$.

Scaling beyond current validation. The Gaussian and particle trackers validate the theory in controlled Gaussian and non-Gaussian settings, and the real-data benchmarks show that the diagnostics remain useful outside the synthetic environment. High-dimensional state spaces, richer action channels, and direct measurements of intervention effort are the next natural tests of the framework.

11 Conclusion

Tracking under drift with finite effective memory obeys a sharp limit. When total error is written as a sum of estimation variance and drift-induced staleness, optimizing over memory yields the cube-root law $\mathcal{E}_{\min} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$ for the estimator class analyzed here. In that sense, tracking under drift is constrained by the same object that makes online compression useful: memory. More memory reduces noise, but it also increases the age of the compressed model.

Action-coupled settings motivate a minimal decomposition into representation, actuation, and adaptation roles $(\mathcal{P}, \mathcal{A}, \Phi)$. In the class of systems studied here, the decomposition separates three experimentally meaningful failure modes and supports bottleneck-sensitive diagnostics. The score measures ordinary coherence, while $\mathcal{C}^E$ corrects the specific regime in which apparent coherence is maintained by intervention rather than by model adaptation.

The empirical results support this structure. The Gaussian tracker recovers the predicted U-curve and the $\zeta^{1/3}$ scaling law in the controlled setting. The particle tracker shows that feedback masking can keep the standard score high even when the underlying model is stale. On passive benchmarks, generic changepoint detectors such as ADWIN remain stronger default alarms. On the logged KuaiRand benchmark, however, the effort-corrected score detects a mode of feedback-induced degradation that the generic baselines do not capture under the same calibration.

Finite-memory tracking under drift obeys a structural coherence-delay trade-off. Closed-loop systems can hide model failure through feedback, and effort-aware diagnostics can surface that regime. How far this decomposition extends beyond the settings studied here remains open.

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A Code Repository

All implementations are publicly available at:

<https://github.com/jovemexausto/coherence-delay-tradeoff>

The repository contains:

- `experiments/run.py` — single entry point for all experiment domains;
- `experiments/gaussian/` and `experiments/particle/` — core experiment logic;
- `experiments/bikes/`, `experiments/elec2/`, and `experiments/kuairand/` — domain-specific benchmarks;
- `REPRODUCIBILITY.md` — setup and run instructions.

All experiments use `uv` for dependency management (`uv sync`) and are reproducible with fixed random seeds as specified in each script.

B KuaiRand Logged Benchmark Protocol

This protocol instantiates an ethically motivated logged benchmark on top of the KuaiRand sequential recommendation logs. The goal is to test whether an effort-corrected coherence score can surface exposure concentration and filter-bubble-like drift.

Data and signals

The primary state signal is `watch_ratio`. Auxiliary feedback signals include `click` and `like`; `dwell`, `follow`, and `share` are reserved for secondary validation when available. For user u at time t , let $P_{u,t}$ denote the inferred preference state, $A_{u,t}$ the exposure policy, and $\Phi_{u,t}$ the composite feedback signal. All user windows use `window_size` = 20, and video tags are binned from `video_features_basic_pure.csv`.

Effort proxy

The effort proxy is defined as the divergence between the current exposure distribution and the healthy random-exposure baseline distribution:

$$E_{u,t} := D_{\text{KL}} \left(\pi_{u,t}^{\text{current}} \parallel \pi_{u,t}^{\text{healthy}} \right).$$

This choice is directional, decomposable, and directly interpretable as the cost of keeping the observed trajectory close to a healthy baseline. In settings where only aggregate exposure counts are available, total variation distance may be used as a fallback, but KL is the preferred default. We use additive smoothing 10^{-6} in the KL computation.

Protocol

The benchmark is executed in three phases.

1. **Healthy.** Exposure follows the random baseline, diversity is unconstrained, and effort remains low.
2. **Bubble.** Exposure becomes progressively concentrated, so that $E_{u,t}$ rises while standard coherence may remain artificially high.
3. **Collapse.** The concentrated exposure regime stops tracking the healthy baseline, generic detectors begin to fire, and the effort-corrected index should lead the baselines.

Evaluation

Evaluation is carried out sequentially within each user session and then batched across users with strict temporal splits. Reported metrics are:

- detection rate of the bubble/collapse phases,
- median detection delay,
- false positives in the healthy phase,
- sensitivity to λ in $\mathcal{C}^E$,
- comparison against ADWIN, PageHinkley, KSWIN, and NoDrift.

All baseline detectors consume the same effort-adjusted scalar signal, $1 - \mathcal{C}^E$, so the comparison is not biased by detector-specific inputs. The baseline hyperparameters are fixed a priori: ADWIN $\delta = 0.03$; PageHinkley $\delta = 0.005$, threshold = 20, $\alpha = 0.9999$; KSWIN window = 30, stat = 10, $\alpha = 0.001$. When enabled, thresholds are calibrated on healthy windows only.

Controls

Required controls include shuffled labels, random phase boundaries, no-coercion ablations, and removal of auxiliary feedback signals. These controls ensure that the benchmark does not reduce to a hindsight-tuned logged-bandit exercise.